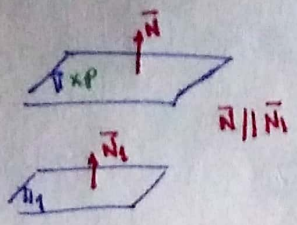
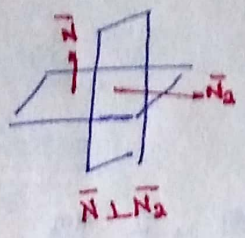
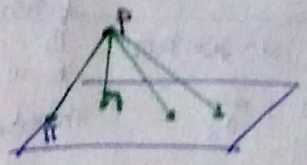






$(a,b) = k(x,y)$

$\boxed{3}$ $\pi: x+y-z=-1$

$P: (1,2,1)$

$P \notin \pi$

$Q \in \pi \mid d(P,Q) = d(P,\pi)$

$L: A=(-1\,2\,2)$ y Q

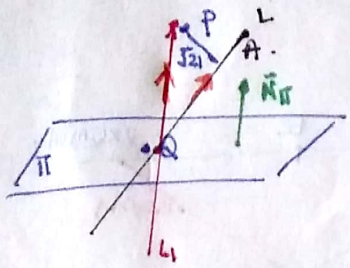
$A \in \pi$.

L

$d(L_P, P) = \sqrt{21}$.

$\downarrow$

d.



$L: \lambda(A-Q) + A.$

$Q = (q_x, q_y, q_z) \Rightarrow L: \lambda((-1,2,2) - (q_x\, q_y\, q_z)) + (-1,2,2)$

$L: \lambda(-1-q_x,\ 2-q_y,\ 2-q_z) + (-1\,2\,2)$

$L_1: \mu(P-Q) + P.$

$L_1: \mu[(1,2,1) - (q_x, q_y\, q_z)] + (1\,2\,1)$

$L_1 = \mu(1-q_x,\ 2-q_y,\ 1-q_z) + (1\,2\,1)$

$L_1 \perp L$

$d_{L_1} \cdot d_L = 0.$

$\downarrow$

$(1-q_x, 2-q_y, 1-q_z)(-1-q_x, 2-q_y, 2-q_z) = 0.$

$d_{L_1} \parallel N_\pi \Rightarrow (1-q_x, 2-q_y, 1-q_z) = k(1\ 1\ -1)$

$1-q_x = 1 \rightarrow q_x = 0.$

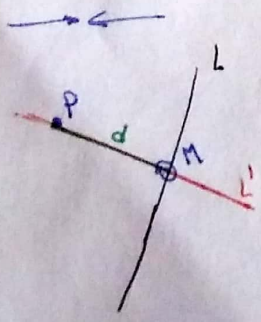
$2-q_y = 1 \rightarrow q_y = 1$

$1-q_z = -1 \rightarrow q_z = 2$

$Q = (0,1,2)$

$L_1: \mu(1\,1\,-1) + (1\,2\,1)$

$L: \lambda(-1, 1, 0) + (-1\,2\,2)$



$M \in L$

$M \in L'$

$L': n(P-M) + P.$

$P = (1\,2\,1)$

$M = (m_x, m_y, m_z)$

$M = (-\lambda-1, \lambda+2, 2) \rightarrow$

$L': n(1+\lambda+1,\ 2-\lambda-2,\ 1-2) + P.$

$L': n(\lambda+2,\ -\lambda,\ -1) + P.$

$d_L \perp d_{L'}$

$d_L \cdot d_{L'} = 0. \rightarrow (-1\,1\,0)(\lambda+2, -\lambda, -1) = 0.$

$-\lambda-2 - \lambda + 0 = 0.$

$-2\lambda - 2 = 0$

$-1 = \lambda \rightarrow M = (0,1,2)$

$d(Q\,P) = \sqrt{21}.$

$\sqrt{x^2+y^2+z^2} = d.$

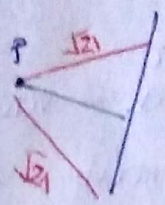
$L: \lambda(d) + P.$

$L: x.A + yB + z.C + a.$

$P = (x_0, y_0, z_0)$

$\cancel{d(P,L) = \frac{|\vec{d}_L \cdot (x_0, y_0, z_0)|}{}}$

$d(P,L) = \left| \frac{A.x_0 + By_0 + Cz_0 + a}{\sqrt{A^2+B^2+C^2}} \right|$



$d(L,P) = \sqrt{21}$

$\sqrt{21} = \sqrt{x^2+y^2+z^2}.$

$21 = x^2+y^2+z^2.$

$21 = (-\lambda-2)^2 + \lambda^2 + 1^2$

$20 = (\lambda^2) + (-\lambda)(-2).2 + (-2)^2 + \lambda^2.$

$20 = \lambda^2 + 4\lambda + 4 + \lambda^2.$

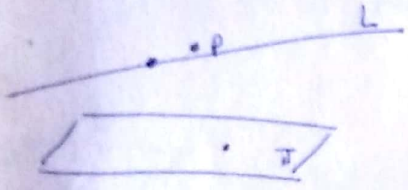
$20 = 2\lambda^2 + 4\lambda + 4 \longleftarrow$ $\lambda_1 \longrightarrow L: M_1$, $\lambda_2 \longrightarrow L: M_2$

$\longrightarrow L: -\lambda-1, \lambda+2, 2$

$P = (1,2,1)$

$L-P: (-\lambda-1-1; \lambda+2-2, 2-1)$

$L-P = \underset{x}{-\lambda-2}, \underset{y}{\lambda}, \underset{z}{1}$



a) $\overset{M}{(x,y,z)}$ $\quad \pi: x_1 + 2x_2 = 0$

$x_1 = -2x_2$

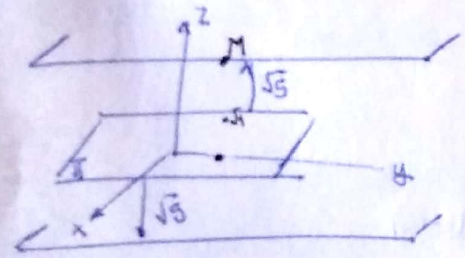
$(-2x_2, x_2, 0)$

$x_2(-2,1,0) = R.$

$d(MR) = \sqrt{5}.$

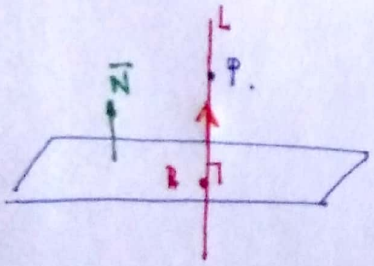
$(x+2)^2 + (y-1)^2 + z^2 = 5.$

$z = \sqrt{5 - (x+2)^2 - (y-1)^2}.$



$M \in \mathbb{R}^3 \,/\, M = (x, y, \sqrt{5-(x+2)^2-(y-1)^2})$

$\cancel{5-4}$ $\sqrt{1}$

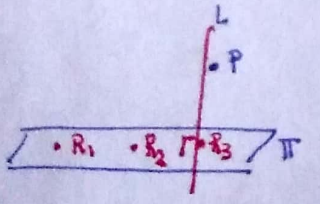


$L: \lambda N_\pi + P. \longrightarrow L: \lambda(x_L, y_L, z_L) + (P_x, P_y, P_z)$ $\quad d_L // N_\pi$

$R \in L \rightarrow R: (\lambda x_L + P_x, \lambda y_L + P_y, \lambda z_L + P_z)$

$R \in \pi \rightarrow$

$A\underset{\lambda x_L + P_x}{x} + B\underset{\lambda y_L + P_y}{y} + C\underset{\lambda z_L + P_z}{z} = d. \longrightarrow \lambda. \longrightarrow R.$



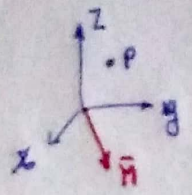
$ax + by + cz = d$

$-x - 3y + 2z = -4$

$a = -1 \quad b = -3 \quad c = 2 \quad d = -4$

$P = (x, y, z)$

(punto)



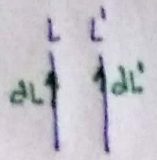
vector: $\bar{M} = (m_x, m_y, m_z)$

sentido, dirección, modulo

Recta: $L: \lambda(P-Q) + P$

$+ Q$

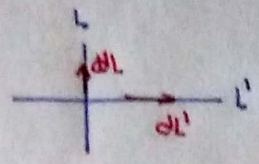
$d_L: P-Q$



$L \parallel L'$

$d_L \parallel d_{L'}$

$d_L = k\, d_{L'}$



$L \perp L'$

$d_L \perp d_{L'}$

$d_L \cdot d_{L'} = 0.$

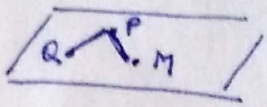
Plano: $\pi: ax + by + cz = d.$

$\bar{N}$ Q $\quad \bar{N}: (n_x, n_y, n_z) = (a, b, c)$

$Q: (q_x, q_y, q_z)$

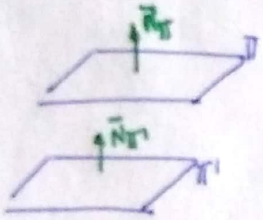
$\bar{N}$ en $\pi \Rightarrow n_x \cdot x + n_y \cdot y + n_z z = d.$

$Q \longrightarrow n_x \cdot q_x + n_y \cdot q_y + n_z q_z = d$



$\pi: [(xyz) - P]\bar{N} = 0.$

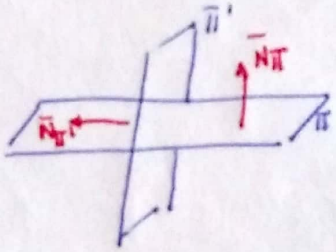
$PQM \rightarrow \bar{N}: (P-Q) \times (Q-M) = (a, b, c)$



$\pi \parallel \pi'$

$\bar{N}_\pi \parallel \bar{N}_{\pi'}$

$\bar{N}_\pi = k\bar{N}_\pi$

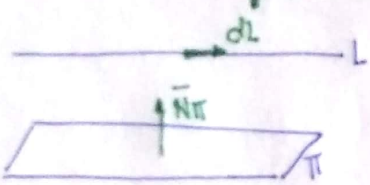


$\pi \perp \pi'$

$\bar{N}_\pi \perp \bar{N}_{\pi'}$

$\bar{N}_\pi \cdot \bar{N}_{\pi'} = 0$

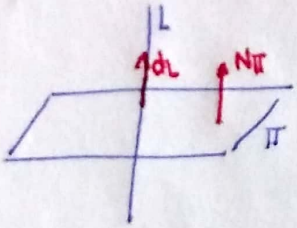
vector director de $L: \lambda\underbrace{(P-Q)}_{d_L} + P$



$\pi \parallel L$

$\bar{N}_\pi \perp \bar{d}_L$

$\bar{N}_\pi \cdot \bar{d}_L = 0$



$\pi \perp L$

$\bar{N}_\pi \parallel \bar{d}_L$

$\bar{N}_\pi = k\,\bar{d}_L$

$\underbrace{\pi_1 \cap \pi_2}_{} \cap \pi_3 = A.$

$x + y = 2x + 3y = 4x + 7y = 9$